EFU MINING LICENCE v1.0

Experimental EFU-based Mining & Quarrying Framework

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0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for mining (metals, coal, aggregates, industrial minerals). NOT a mining concession, environmental impact assessment, or worker safety permit replacement, but an open research protocol measuring mining's physical footprint in human-scale EFU metrics (energy, materials, tailings, land disruption). Covers surface, underground, alluvial, and quarrying operations.library

I. Core Concepts

EFU-Mining (EFU-MINING)
EFU framework application for extractive industries:

- EFU-E_MIN: energy (diesel, electricity for drilling/blasting)
- EFU-M_MIN: explosives, steel wear, machinery parts
- EFU-W_MIN: water (processing, dust suppression, tailings)
- EFU-T_MIN: tailings/waste rock volume
- EFU-LAND_MIN: land disturbance + topsoil loss

Mining Sovereignty Gap (SS_MINING)
SS_MINING = TEFU_MINING_total – Regional Mineral Carrying Capacity (RMCC_MINING)
Measures if extraction stays within regional sustainable extraction capacity (50-100 year planning horizon).

MROI_MINING
MROI_MINING = THI_MINING / TEFU_MINING
THI_MINING = critical materials supply + economic value + recycling content.library

II. Licence Status

Documents EFU-audited mining operations, quarries, processing plants. Enables copper mine vs. sand quarry benchmarking. Compatible with ISO 14001 environmental management systems.library

III. Metrics and Thresholds

1. EFU Components (per tonne extracted metal/mineral)

Component	Formula	Example (Copper)
EFU-E_MIN	energy_MWh / 12.88 / tonne	40 MWh/t → 0.0031 EFU/t
EFU-M_MIN	explosives_steel_kg / 30kg/person/year / tonne	-
EFU-W_MIN	water_m³ / 1000m³/person/year / tonne	-
EFU-T_MIN	tailings_m³ / 100m³/person/year / tonne	-
TEFU_MINING = 1.3×EFU-E + 1.4×EFU-M + 1.2×EFU-W + 2.0×EFU-T + 1.5×EFU-LANDlibrary		

2. Regional Mineral Carrying Capacity (RMCC_MINING)
RMCC_MINING = Sustainable_extraction_kt / Population / 0.1t/person/year

IV. Licence Categories

Level	SS_MINING	MROI_MINING	Label	Interpretation
1	≤ 0	≥1e-2	EFU-MINING SOVEREIGN	Within capacity
2	0 < ≤20k EFU	≥5e-3	EFU-MINING BALANCED	Sustainable reserves
3	20k < ≤100k EFU	≥1e-3	EFU-MINING DEFICIT	Import dependency
4	>100k EFU	<1e-3	EFU-MINING CRITICAL	Exhaustion risk

V. Regional Calibrations

Country	Extraction (Mt/year)	Population (M)	RMCC_MINING (EFU/person/year)
AUSTRALIA	500	26	0.19
CHILE	200	19	0.11
CHINA	10,000	1,410	0.007
USA	2,000	340	0.006
EU	3,000	448	0.007

VI. Example: Australian Iron Ore (BHP/Rio Tinto)

Annual Sector Input: 900Mt ore, 45B MWh energy, 5Mt explosives/steel, 1.2B m³ tailings
Calculations:

- EFU-E_MIN = 45B/12.88/900M = 0.0039 EFU/t
- EFU-M_MIN = 5M/30/900M = 0.00019 EFU/t

- EFU-T_MIN = 1.2B/100/900M = 0.0133 EFU/t
  TEFU_MINING_total ≈ 12,500,000 EFU/year
  Allocated (national) = 26M × 0.19 = 4,940,000 EFU/year
  SS_MINING = 12.5M – 4.94M = +7,560,000 EFU/year (Level 4: CRITICAL)
  Result: EFU-MINING CRITICAL (Aggregates/sand = Level 2: BALANCED).library

VII. Implementation Notes

- **Critical minerals**: TEFU_MINING prioritizes green transition materials
- **Tailings regulation**: EFU-T_MIN assesses dam failure risk
- **Reclamation**: EFU-LAND_MIN basis for restoration bonds
- **Data sources**: Tier 1 = fuel receipts/blast logs; Tier 2 = USGS production data

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